

CUAS Decision Map — User Guide

A map-based decision-support tool for choosing a counter-UAS (counter-drone) countermeasure at a specific location — what's legally permitted, how risky it is, and what to file when you can't act yourself.

Pilot: Texas

Federal + state law

Live airspace & aircraft

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1 · The screen at a glance

The app is a single screen: a control sidebar on the left, an interactive map on the right. You pick *who you are* (your authority), then *click a location*. The tool reads that location's airspace, population, terrain, and live air traffic and returns a tailored decision report.

CUAS Decision Map

Click the map to assess - pilot: Texas

AUTHORITY PROFILE

Federal — DHS/DOJ (6 U.S.C. § 124n) ▾

☒ Credible threat declared

MAP LAYERS

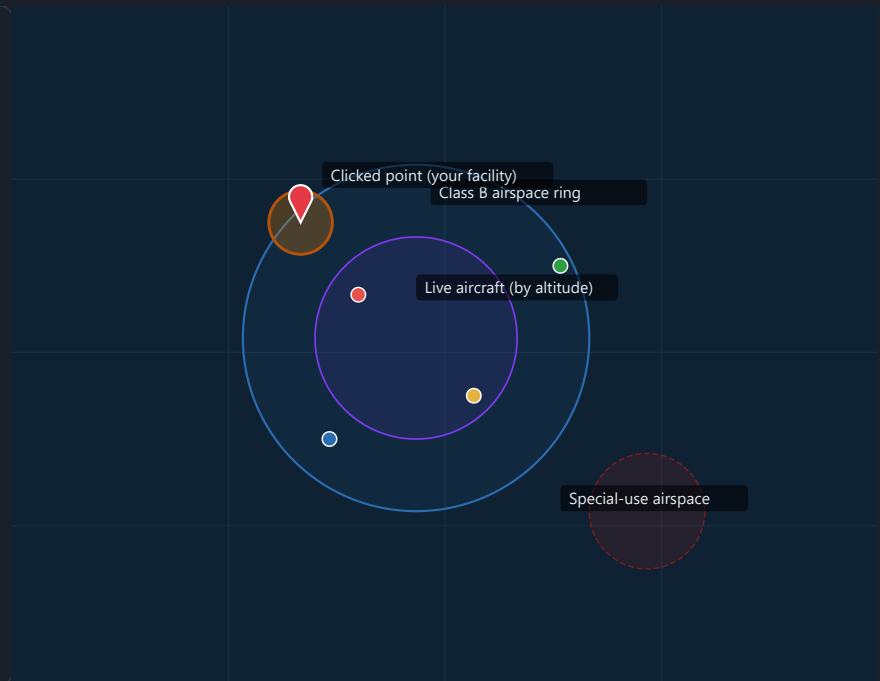
- ☒ Live aircraft (ADS-B)
- ☒ Airspace (FAA Class B/C/D)
- ☒ Special-use airspace
- ☒ Population density
- ☒ Restricted zones

DHS/DOJ covered-facility authority:
detection + mitigation w/ FAA coordination

Permitted 4

Conditional 8

Prohibited 0



The full screen: authority profile + layer toggles + decision report on the left; airspace, zones, special-use airspace, and live aircraft on the map.

2 · Authority profile — the master filter

This is the most important control. **Who you are determines what the law lets you do**, before any location analysis. Pick your profile and the entire report adapts.

Federal — DHS/DOJ (6 U.S.C. § 124n)

Detect + mitigate credible threats with FAA coordination.

Federal — DoD (10 U.S.C. § 130i)

Covered military installations.

Federal — DOE/NNSA (FY26 NDAA § 3111)

Nuclear-security facilities.

State/Local LE — certified (SAFER SKIES)

Limited mitigation with DOJ certification. GPS-jam, HPM, HEL stay off-limits.

State/Local LE — not certified

Detection only; pursue certification.

Critical-infrastructure owner / Private security

Detection only; mitigation is a federal matter — see the documentation pathway.

The same location returns very different answers by profile — a federal team sees a ranked menu of mitigation options; a private operator sees detection plus "here's what to file and where."

3 · Map layers

Toggle layers on the left. Each one feeds the assessment — they're not just decoration.

- FAA Class B / C / D airspace (real)
- Special-use airspace (Prohibited / Restricted / MOA)
- Restricted zones (critical infrastructure, stadium, prison)
- Population density (drives debris-risk scoring)

LIVE AIRCRAFT — colored by altitude

- low / overhead
- ~1,500 ft
- ~5,000 ft
- high

Red = low & overhead = highest collateral concern for kinetic options.
Aircraft refresh every ~15 s and when you pan the map.

Source: FAA ArcGIS (airspace) · adsb.lol / airplanes.live (aircraft).

Population, buildings, and bespoke zones are sample data in this pilot; airspace and aircraft are live.

Layer legend. Airspace and aircraft are live feeds; population/zones are pilot sample data pending the next data swap.

4 · The decision report

Click any point and the sidebar fills with a report in four parts: an **authority banner**, then countermeasures grouped as **Permitted** **Conditional** **Prohibited**, then the documentation pathway. Actionable options are ranked safest-first.

Certified SLTT authority (SAFER SKIES): limited mitigation of credible threats with DOJ certification + FAA coordination. GPS jam/spoof, HPM, and HEL remain OUTSIDE this authority.

Dallas Class B · airspace B · nearest DFW (0.0 nm) · pop. high · 12 aircraft nearby

Permitted 4

RF detection / direction-finding

Passive; do not decode the control link (Wiretap/Pen-Trap).

risk 12 · low

Conditional — requires the listed conditions · 5

Kinetic (projectiles / firearms)

risk 100 · high

Conditions: credible_threat, faa_coordination, doj_certification, clear_fall_zone
Drivers: high-density area · manned aircraft 0.9 nm @ 525 ft · within 0.0 nm of DFW
18 U.S.C. § 32

Prohibited 3

GNSS/GPS jamming or spoofing

Not in the SLTT pathway; uncontrollable aviation effects. 47 U.S.C. § 333; 49 U.S.C. § 46307

High-power microwave (HPM)

Federal/military only. 47 U.S.C. § 302a; § 333

Documentation & coordination pathway

Establish real-time FAA coordination

Authorized mitigation requires FAA deconfliction before action.

A decision report for a certified state/local LE profile at DFW. Note the SAFER-SKIES carve-out (GPS-jam/HPM stay prohibited) and the live-aircraft risk driver.

Every countermeasure card carries its **legal citation** as a clickable chip (e.g. [18 U.S.C. § 32](#)), its **conditions**, and a **risk score**. The footer stamps the rules-database version and date — because the law changes.

5 • Collateral-risk scoring

Each actionable countermeasure gets a 0–100 collateral-risk score and a band, with the top drivers shown. The model reflects how each method actually fails at this location:

Kinetic / net / interceptor — falling-debris & ballistic fall-zone vs. population density; a manned aircraft within 5 nm and below 1,000 ft pushes risk sharply up.

RF jamming / HPM — interference radius vs. nearby cellular, GPS, Wi-Fi and public-safety comms; worse in dense RF environments and near airports.

GNSS jam/spoof — near-always high: GPS effects can't be contained and hit aviation and timing.

Detection (RF / radar / EO-IR / acoustic) — minimal collateral, scored low.

low 0–33

medium 34–66

high 67–100

Transparent heuristics for decision support — not engineering models.

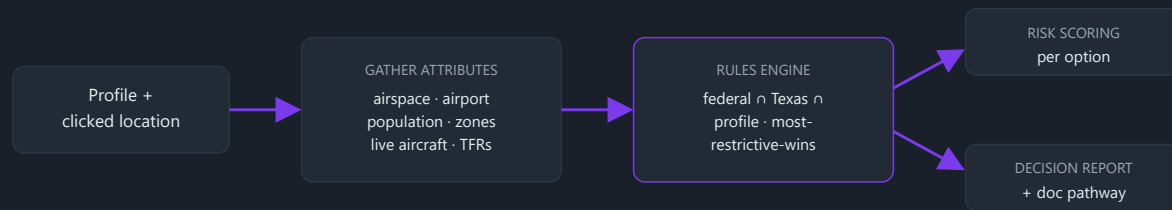
6 • Documentation & coordination pathway

When you can't lawfully act, the most useful output is **what to file and where**. The tool assembles the pathway from your profile and the location:

- **Report the incursion** — FBI field office / DHS, so authorized teams can respond.
- **FAA § 2209 fixed-site petition** (proposed Part 74) — establish a UAS no-fly designation over your facility.
- **§ 99.7 Special Security Instruction** — for national-security-sensitive sites, via your sponsoring federal agency.
- **SAFER SKIES certification** — the path for state/local LE to gain lawful mitigation authority.
- **FEMA C-UAS grant** — federal funding for approved capability.
- **Texas DPS / iWatch Texas** — when the drone is over Texas critical infrastructure (Gov't Code ch. 423).

Each item links to the authoritative source.

7 · How a decision is made



The pipeline behind every click: gather location attributes → apply the cited rules database → score risk → assemble the report.

8 · What it is — and isn't

It is

A planning / course-of-action aid that makes a complex, multi-variable legal and safety judgment faster and more defensible — with citations and an audit trail.

It isn't

An authorization system or a real-time fire-control tool. Live aircraft are situational awareness, not a clearance to engage.

Pilot scope: Texas, federal + Tex. Gov't Code ch. 423. Live data: FAA airspace, community ADS-B. Sample data: population, buildings, bespoke zones.